

MATH 2211, SPRING 2026: SOLUTIONS TO HOMEWORK 8

- (1) Show that the following are equivalent for a matrix $A \in M_{n \times n}(F)$ and a scalar $\lambda \in F$:
- (a) There exists a non-zero vector $v \in F^n$ such that $Av = \lambda v$.
 - (b) The RREF of $A - \lambda I_n$ has a free variable.
 - (c) If $T : F^n \xrightarrow{v \mapsto Av} F^n$ is the linear transformation associated with A , then $T - \lambda \cdot \text{id}_{F^n}$ is not injective.

Solution. All assertions are equivalent to saying that $\ker(T - \lambda \text{id})$ is non-zero. See 2]lecture22Observation in Lecture 22 for the 2×2 case.

- (2) Prove that the determinant of 2×2 -matrices has the following property:

$$\det(AB) = \det(A) \det(B)$$

for all $A, B \in M_2(F)$.

Solution. This is a direct computation: If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$, then $AB = \begin{bmatrix} ap + bq & ar + bs \\ cp + dq & cr + ds \end{bmatrix}$, and so

$$\begin{aligned} \det(AB) &= (ap + bq)(cr + ds) - (ar + bs)(cp + dq) \\ &= acpr + bcqr + adps + bdqs - acpr - bcps - adqr - bdqs \\ &= bcqr + adps - bcps - adqr \\ &= (ad - bc)(ps - qr) \\ &= \det(A) \det(B). \end{aligned}$$

We will see in class that there is a *unique* function $\det : M_n(F) \rightarrow F$ with the following properties:

- (a) (Identity) $\det(I_n) = 1$.

(b) (Alternating property) $\det \begin{pmatrix} v^T \\ v^T \\ \vdots \\ v_n^T \end{pmatrix} = 0;$

(c) (Scaling) $\det \begin{pmatrix} c \cdot v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} = c \cdot \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix}$

(d) (Addition) $\det \begin{pmatrix} v_1^T + w_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} = \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix} + \det \begin{pmatrix} w_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{pmatrix}$

- (3) Show that $\det$ also has the following properties:

$$\begin{aligned}
 \text{(a) (Swapping property)} \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{pmatrix} &= -\det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_{i+1}^T \\ v_i^T \\ \vdots \\ v_n^T \end{pmatrix} \\
 \text{(b) (Adding rows property)} \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{pmatrix} &= \det \begin{pmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_i^T + v_{i+1}^T \\ v_{i+1}^T \\ \vdots \\ v_n^T \end{pmatrix}
 \end{aligned}$$

- (4) For any matrix $B \in M_n(F)$, show that the function $A \mapsto \det(AB)$ also has the defining properties (b), (c), (d) of $\det$, except that it takes I_n to $\det(B)$. Use this to show that we have $\det(AB) = \det(A) \det(B)$ even in the $n \times n$ -case.

Solution. Write $A = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}$, $B = [u_1 \ u_2 \ \cdots \ u_n]$. The i -th row of AB now looks like

$$(0.0.0.1) \quad [v_i \cdot u_1 \quad v_i \cdot u_2 \quad \cdots \quad v_i \cdot u_n]$$

Now, let's look at each of the properties in turn:

- (Alternating property) If $v_1^T = v_2^T = v^T$, then we see that the first and second rows of AB are also equal, and so $\det(AB) = 0$.
- (Addition) The formula in (0.0.0.1) tells us that when the first row looks like $v_1^T + w_1^T$, then the first row of AB looks like

$$[v_1 \cdot u_1 \quad v_1 \cdot u_2 \quad \cdots \quad v_1 \cdot u_n] + [w_1 \cdot u_1 \quad w_1 \cdot u_2 \quad \cdots \quad w_1 \cdot u_n]$$

and so this is also a sum. Applying the addition formula for $\det$ now tells us that $\det(AB)$ is also a sum of two determinants, each controlled by one of the summands in the first row.

- (Scaling) This is easy, since (0.0.0.1) shows that the first row of AB also gets scaled by the same quantity k , and so the determinant also gets scaled by k .

Now, there are two possibilities (see6]lecture24Lecture 24):

- B is invertible and $\det(B) \neq 0$.
- B is non-invertible and $\det(B) = 0$.

In the first case, the function $A \mapsto \det(AB) \det(B)^{-1}$ now satisfies *all* the properties satisfied by $\det$, and so must be *equal* to it. In other words, we must have

$$\det(A) = \det(AB) \det(B)^{-1} \Rightarrow \det(A) \det(B) = \det(AB).$$

In the second case, $\det(A) \det(B) = 0$, but also $\det(AB) = 0$, because AB is also non-invertible. You can see this directly or appeal to Homework 7, Problem 1. Therefore, $\det(AB) = 0 = \det(A) \det(B)$.

(5) Use row operations to compute $\det(A)$ where

$$A = \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 2 & 5 & 0 & 2 & 3 \\ -1 & -2 & 4 & 1 & 0 \\ 0 & 3 & 6 & 11 & 5 \\ 1 & 4 & 1 & 5 & 7 \end{bmatrix}$$

Solution. Let us run through it.

$$\begin{aligned} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 2 & 5 & 0 & 2 & 3 \\ -1 & -2 & 4 & 1 & 0 \\ 0 & 3 & 6 & 11 & 5 \\ 1 & 4 & 1 & 5 & 7 \end{bmatrix} &\xrightarrow[R5-R1]{R2-2R1, R3+R1} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 3 & 1 & 1 \\ 0 & 3 & 6 & 11 & 5 \\ 0 & 2 & 2 & 5 & 6 \end{bmatrix} \\ &\xrightarrow[R4-3R2]{R5-2R2} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 3 & 1 & 1 \\ 0 & 0 & 0 & 5 & 2 \\ 0 & 0 & -2 & 1 & 4 \end{bmatrix} \\ &\xrightarrow[R5+\frac{2}{3}R3]{R5-\frac{1}{3}R4} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 3 & 1 & 1 \\ 0 & 0 & 0 & 5 & 2 \\ 0 & 0 & 0 & \frac{5}{3} & \frac{14}{3} \end{bmatrix} \\ &\xrightarrow{R5-\frac{1}{3}R4} \begin{bmatrix} 1 & 2 & -1 & 0 & 1 \\ 0 & 1 & 2 & 2 & 1 \\ 0 & 0 & 3 & 1 & 1 \\ 0 & 0 & 0 & 5 & 2 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix} \end{aligned}$$

One could continue, but at this point, we can directly calculate the determinant as the product of the diagonal entries (see 9]lecture26Lecture 26 for the 3×3 case). This is because the remaining row operations that we have to do are all of the form where we add a multiple of one row to a different row.

In any case, all the operations we performed above are of this form, and so we never changed the determinant! So the determinant of the original matrix is just $3 \cdot 5 \cdot 4 = 60$.